

Numerical simulation of the dual-core structure of the Bohai Sea cold bottom water in summer

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Regional Ocean Modeling Systems (ROMS 3.0) and the κ - ϵ turbulence closure scheme has been applied to investigating the seasonal evolution of the thermocline in the Bohai Sea. The simulation reproduces the stratifications lasting from early April to early September and reveals the existence of marked Asymmetric Dual-Core Cold Bottom Water (ADCCBW) in the south and north depression basin respectively under the thermocline. The bottom temperature in the north depression is about 1–4°C lower than that in the south depression basin which is in good agreement with observations. Model results suggest that the local bathymetry characteristics and inhomogeneous net heat flux due to the latitude difference are the major cause for the early formation of the ADCCBW. Numerical Lagrangian drifter experiments support the finding that the ADCCBW is maintained throughout the stratification periods by the inflow of cold bottom water from the northern Yellow Sea and deep channel in the western side of Liaodong Peninsula. The inflow cold water contributes to the north depression basin distinctively larger than to the south one. Tidal mixing enhances the bottom temperature asymmetry between the two basins.

Bohai Sea, stratification, ROMS, ADCCBW, depression basin

The Bohai Sea is a semi-enclosed shallow sea, with an average depth of 18 m, consisting of the Bohai Bay, the Laizhou Bay, the Liaodong Bay (LDB) and the central bank, connecting with the Yellow Sea through the Bohai Strait in the east. The central bank is shallower than 25 m surrounded by two depression basins which are deeper than 25 m in the south and 30 m in the north respectively. Thus the bathymetry along the transect from the mouth of the Yellow River (MYR) to the north of Liaodong Bay, is like an asymmetric letter “w” (Figure 1).

The temperature distribution in the Bohai Sea features in accordance with the above-mentioned bathymetric characteristics. It is commonly accepted that the Bohai Sea stratifies from early April to early September. Under the seasonal thermocline, there are two separate cold water masses centered around the two depression basins in which the bottom temperature is usually not equal to

each other; this may be referred to as Asymmetric Dual Core Cold Bottom Water (ADCCBW). This structure can be recognized by the multi-year hydrological atlas^[1] (Figure 2), and *in situ* data sampled in June and July 1959^[2], in late August and early September 2000^[3]. The temperature difference in the two basins is about 1°C in July and August and 1.5°C in later summer according to historical observations^[2,3]. The transition between the cold core and well mixed shallow coast warm water are thermal fronts dominated by tidal mixing^[2,4–6].

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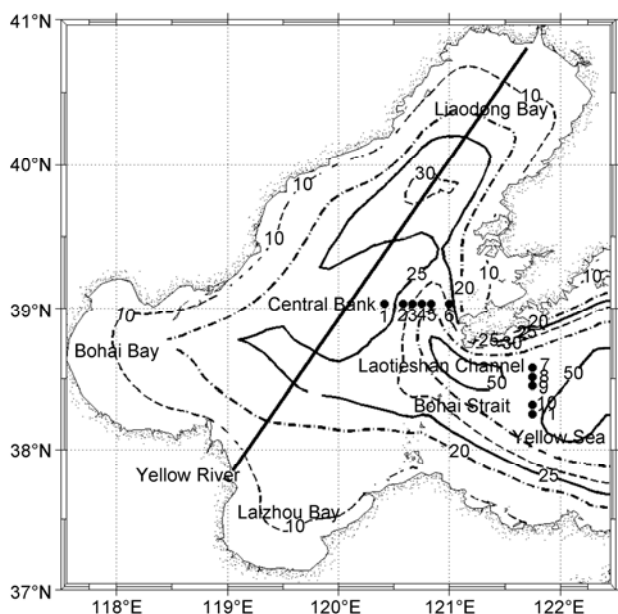


Figure 1 Bathymetry of the Bohai Sea and model domain. Thick slant line denotes the transect from MYR to LDB. Solid circles denote releasing locations of Lagrangian drifters, see section 2.3.

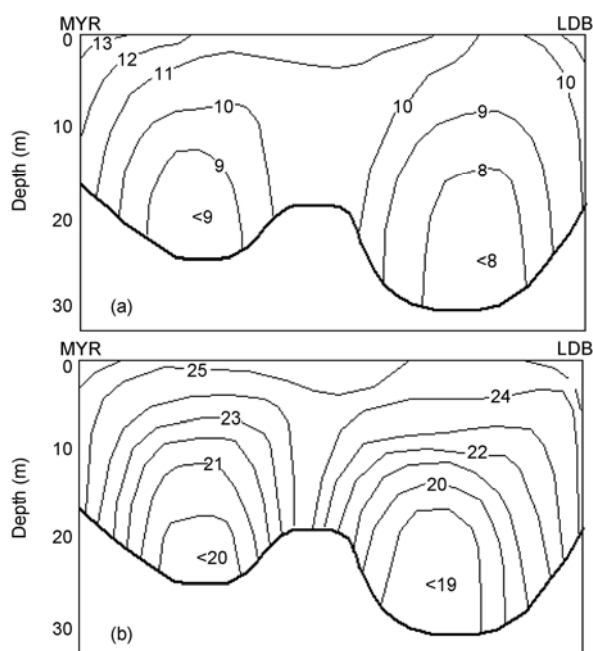


Figure 2 Climatological temperature profile along the transect from MYR to LDB specified in Figure 1: (a) May; (b) August (modified from ref. [11]).

The formation and evolution of the seasonal thermocline in the Bohai Sea have been intensively studied in recent decades. Huang et al. [4,5] have applied a Z-coordinated Hamburg Shelf Ocean Model (HAMSOM) to investigate the 3D baroclinic circulation and tidal fronts;

they suggest that the surface cold water belt along the tidal fronts in July is induced by entrainments of the bottom water under the strong thermocline. But the cold bottom water in their study has only a single core due to the coarse bathymetry and vertical resolution in their model. Liu et al. [6] used the ECOM model and Mellor-Yamada 2.5 turbulence closure scheme to study the M_2 tide and its density residual currents with a more realistic geometry and higher resolution, and obtained the ADCCBW structure. But the vertical mixing in their model is believed to be over-estimated so that the stratification was unrealistically reduced in summer. Lin et al. [7] used POM and also Mellor-Yamada 2.5 turbulence closure scheme, considering tidal mixing as well as surface wave enhanced mixing, and recreated the dual-core cold water structure and more reasonable thermocline above the two bottom cores. However, the temperature in the north basin was almost equal to that in the south basin in all experiments of their simulations. Wu et al. [8] referred to previous simulation studies have not yet reached a certain consensus on the structure of the circulation in the Bohai Sea.

To gain a better understanding of the generation and preservation of the ADCCBW, we use a 3D numerical model with realistic bottom topography to examine the distribution of ADCCBW and seasonal variations of the thermocline and circulation in the Bohai Sea.

1 Model and data

The numerical model applied here is the Regional Ocean Modeling Systems v3.0 (see <https://www.myroms.org/>). ROMS [9–13] is a widely used free surface, hydrostatic, primitive equation ocean circulation model in nonlinear terrain-following coordinates (S-coordinates) [14], which is based on modular design and supplies different kinds of discrete schemes for the momentum and tracer equations and several options for turbulence closure schemes.

The model region covers the whole Bohai Sea with an extension eastward into the northern Yellow Sea where the only open boundary is set at section $122^{\circ}30'E$ which is far away from the strait (Figure 1). The model uses a curvilinear grid with an average horizontal resolution of $1^{\circ}/12'$ in each dimension. There are 15 vertical levels placed and the surface and bottom S-coordinates control parameters are set to 5.0 and 0.4 respectively so that both of the bottom and top layers can be well resolved

(Figure 3). The κ - ε scheme is selected to calculate the sub-grid vertical mixing coefficients as Warner et al. [13] described under the integrative framework of a general length-scale equation proposed by Umlauf and Burckard [15]. The background values for momentum and tracers are set at $1.0 \times 10^{-5} \text{ m}^2/\text{s}$ and $1.0 \times 10^{-6} \text{ m}^2/\text{s}$ respectively. The horizontal mixing was simulated along geopotential surfaces with initial coefficients of $20 \text{ m}^2/\text{s}$ and $10 \text{ m}^2/\text{s}$ for momentum and tracers.

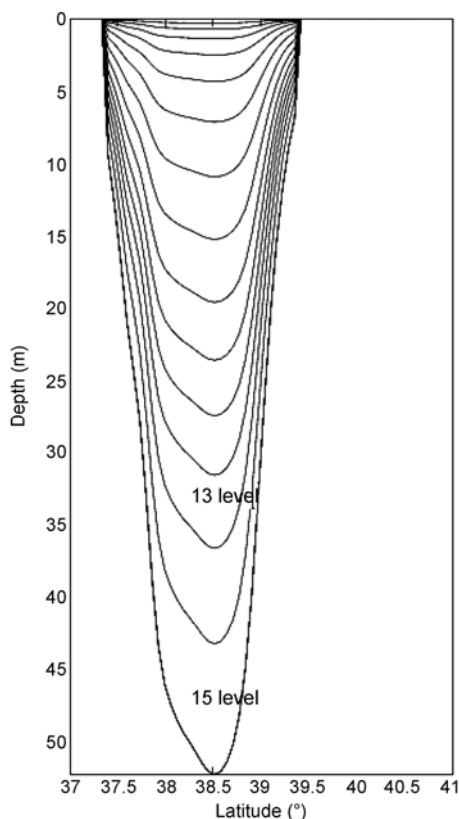


Figure 3 Model vertical levels in the east boundary.

The model is initialized from a climatological temperature and salinity from multi-year hydrologic atlas [1]. The surface boundary is imposed by climatological monthly wind forcing, heat and fresh water flux using the Comprehensive Ocean-Atmosphere Data Set (COADS) [16] with 1° resolution. At the open boundary both of temperature and salinity are linearly interpolated from the multi-year hydrologic atlas [1]. A multi-year average velocity is prescribed derived from monthly Simple Ocean Data Assimilation (SODA) reanalysis from 1958 to 1979 [17,18].

The most dominant tidal constituent in the region, M_2 , is taken into account. The model time step is 621 s, re-

sulting in 72 time steps over an M_2 tidal cycle (12.42 h). In order to supply the M_2 tidal elevation and phase, a large domain model using HAMSON and covering the region of $117^\circ 30' - 131^\circ \text{E}$, $24^\circ - 41^\circ \text{N}$, is run at first for 35 d [19,20], driven by elevations derived from the CSR3.0 global tidal model [21]. The open boundary uses Flather conditions [22] if M_2 is considered or radiations conditions if not. The model achieves dynamic equilibrium after 3 years continuous running. Results taken from the last year are presented in the following section.

A couple of *in-situ* observations are cited to identify the ADCCBW structure and validate the model results. They are: (1) multi-year hydrologic atlas of the Bohai Sea, Yellow Sea and East China Sea [1], (2) interdisciplinary atlas of the Bohai Sea in August of 1982 and 1992 [23], and (3) investigations from cruises in June and July of 1959 [2], from August 29 to September 5 in 2000 and from January 1 to 15 in 2001 [3].

Figure 4 gives the Root Mean Square error (RMS error) by SST comparisons between the simulations and multi-year hydrologic atlas [1]. The average RMS error is as large as 1.7°C over the whole domain caused by the mismatch between the reality and the observations and even false forcing records at the land-sea edge area, but it is only 1.2°C or 1.0°C without regard to the coastal area shallower than 10 m or 15 m.

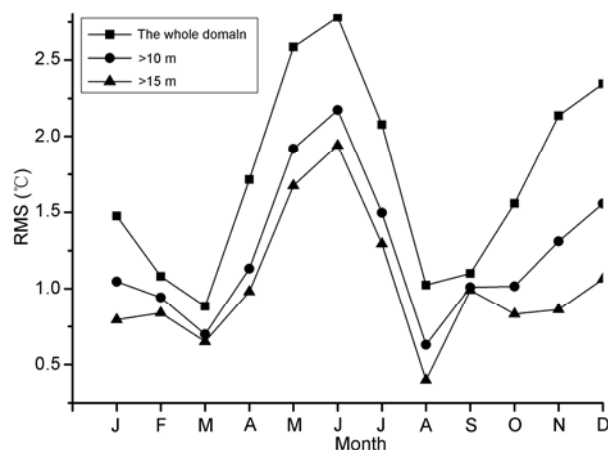


Figure 4 SST RMS error between the 3rd year simulated results and multi-year hydrologic atlas of ref. [1].

2 Results

2.1 Influence of tidal mixing

We first compare M_2 tidal case results with the case without the tidal component to check the tidal mixing

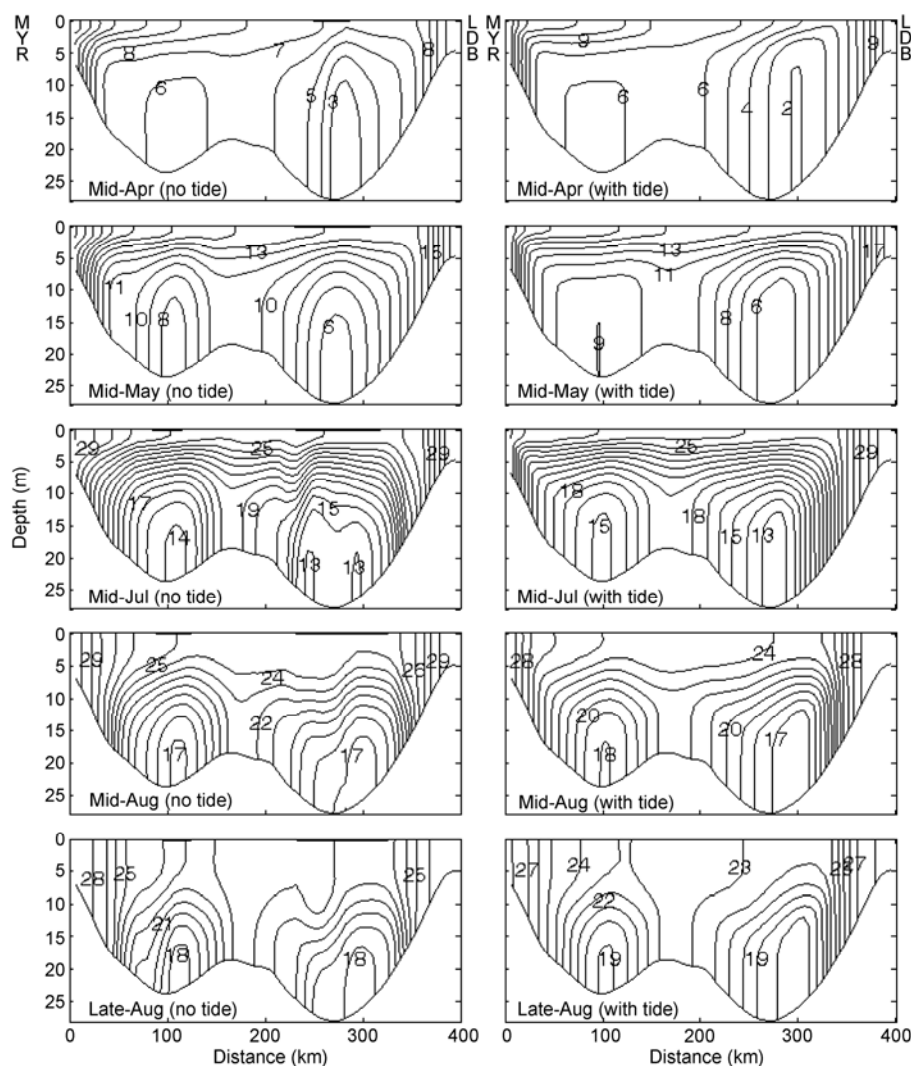


Figure 5 Temperature profile comparisons between no-tide case and tidal case along the transect from MYR to LDB as specified in Figure 1. Left panels are results from no-tide case; right panels are tidal case results.

effects. The temperature along the transect from MYR to LDB (see Figure 1) is shown in Figure 5, in which left panels are 10-day mean experiments without tide from mid-April to late August, all right panels are results with M_2 within the same periods. The results show that the temperature structure and value are rather different between the two cases. The tidal mixing impacts on the two depression basins are not equal, either.

During early periods of stratification in mid-April, the temperatures in the two cases in the south depression are basically similar to each other. But in the north depression, there is 4°C disparity. The temperature in the tidal case is relatively low, capped by a relatively shallow and flat thermocline. In mid-May, the coastal water near the south depression in the no-tide case warms up much faster than

in the tidal case, but the trend in the center of the depression between these two cases is opposite because the M_2 tide reinforces horizontal mixing and heat exchange. The situation in LDB and northern depression differs greatly from the southern Bohai Sea. The only small change caused by tidal mixing is the volume of cold water getting larger. In the following mid-July and mid-August, the role of the tides is no different from those of previous periods. In a word, there is 1°C difference between two cases from May to August in the south depression. However, the tides make only small changes to the temperature structure in the north depression without modifying the minimum value. In the above experiments, the well mixed coastal zone in LDB is wider than in adjacent areas to MYR due to the relatively gentle bottom slope there. So

tidal mixing enlarges the well mixed zone in LDB more significantly than in MYR.

2.2 Evolution of the thermocline

According to the comparison, the tidal case is much more realistic than the other. So the following description of the evolution of the thermocline is based on the tidal case results.

The stratification starts at first on the upper layer of two the basins in mid-April when the coastal region is still well mixed. Under the thermocline, the temperature of the water in the south depression is 6°C in contrast to only 2°C in the north depression. In mid-May, the temperature of the two depressions increases to 9°C and 6°C respectively. They then increase to 15°C and 13°C in mid-July in agreement with observations in the same periods of 1959^[2] when the stratification reaches the strongest and thickest stage overall the year. In mid-August, the temperature rises to 18°C and 17°C respectively. The near-coastal zone all over the Bohai Sea warms as well; however, the thermocline thickness and gradient reduce as the upper mixed layer deepens. The bottom temperature disparity between the two depressions diminishes gradually during these periods. In late August, the water column in the middle bank changes into the homogeneous state. Moreover, the thermocline above the two cold cores reduces significantly, while the two cold cores still increased to 19°C and 18°C.

2.3 ADCCBW maintenances

Figure 6 shows the simulated surface and bottom temperature in mid-August in the Bohai Sea. The SST outside the strait and in the eastern part of LDB is lower than in the other parts (Figure 6(a)), which is close to observations in the historical literature^[2,3]. At the bottom, there is a remarkable cold water tongue extending westward from the northern Yellow Sea in the north side of the strait (Figure 6(b)). The cold tongue bifurcates into two parts to the east of the central bank: the main part turns to the north and has a trend of extending into the north depression, while the rest continues on westward extension into the south depression. The temperature on the banks is relatively high in a closed region like 21°C and 22°C contours.

It is thus suggested that there are potential sources to maintain the cold bottom water in the Bohai Sea during

mature stratification periods. That is, the intruding cold water from the northern Yellow Sea might contribute to preserving the ADCCBW. Huang et al.^[24] and Lin et al.^[25] have studied the flow structure in the Bohai Straits and suggest that there is a basic ‘inflow in the north and outflow in the south’ flow pattern in the strait in all seasons. The pattern of bottom temperature in Figure 6(b) also supports that hypothesis.

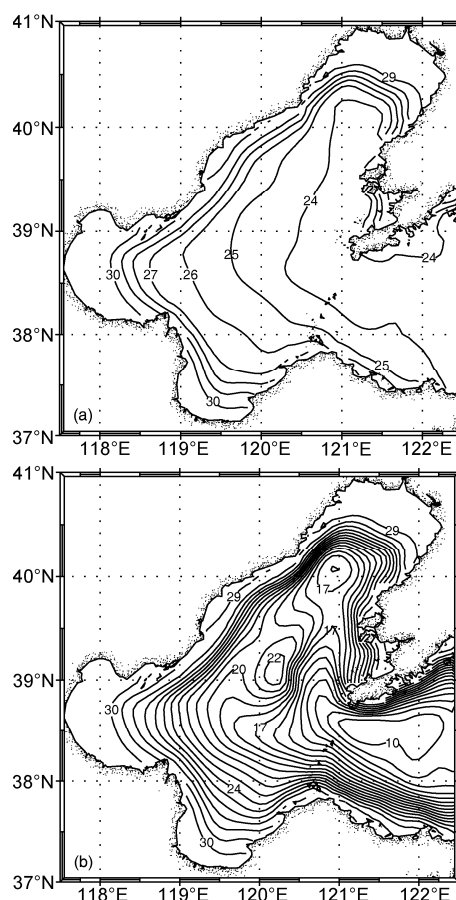


Figure 6 Simulated SST (a) and bottom temperature (b) in mid-August.

To testify the hypothesis, a series of Lagrangian drift-er experiments are designed through the “Random Walk” module provided by ROMS to track their trajectories including diffusion effects. The neutralized floats are placed in 11 locations at the 39°N section (in the north of Laotieshan Channel) and 121°45'E section (in the east of Laotieshan Channel) (Figure 1). Particles are released at the 13th level (sub-bottom or near-bottom) and the 15th level (bottom) in each location, and they are released at the same time at flood tides (zero elevation), high slack tides, ebb tides (zero elevation) and low

slack tides respectively. The releasing time is specified according to sea surface elevations of the 8th site at the 121°45'E section (Figure 1) on May 13, when most of the Bohai Sea are stratified. The tracking is stopped on September 4 when most of the Bohai Sea is un-stratified. The following are focused on trajectories of those released at flood tides and ebb tides since the other two are very similar to them according to the simulated results. The tidal average depth of the 11 sites is listed in Table 1.

Table 1 Tidal average depth of sub-bottom and bottom in float experiments (unit: m)

Float No.	1	2	3	4	5	6	7	8	9	10	11
Sub-bottom level	14	17	19	21	23	18	29	29	30	30	30
Bottom level	23	27	30	34	37	28	47	48	48	48	49
Total depth	23	27	30	34	37	28	47	48	48	48	49

To make a clear description about particle destinations, the northern LDB is used to stand for the region to the north of 39°N section.

The trajectories of floats released in the sub-bottom at the 121°45'E section at flooding periods are shown in Figure 7(a). They all only arrive at the 39°N section after turning around the corner of Liaodong Peninsula and most of them are trapped in the north of the deep Laotieshan Channel except floats 7 and 8. By contrast, both

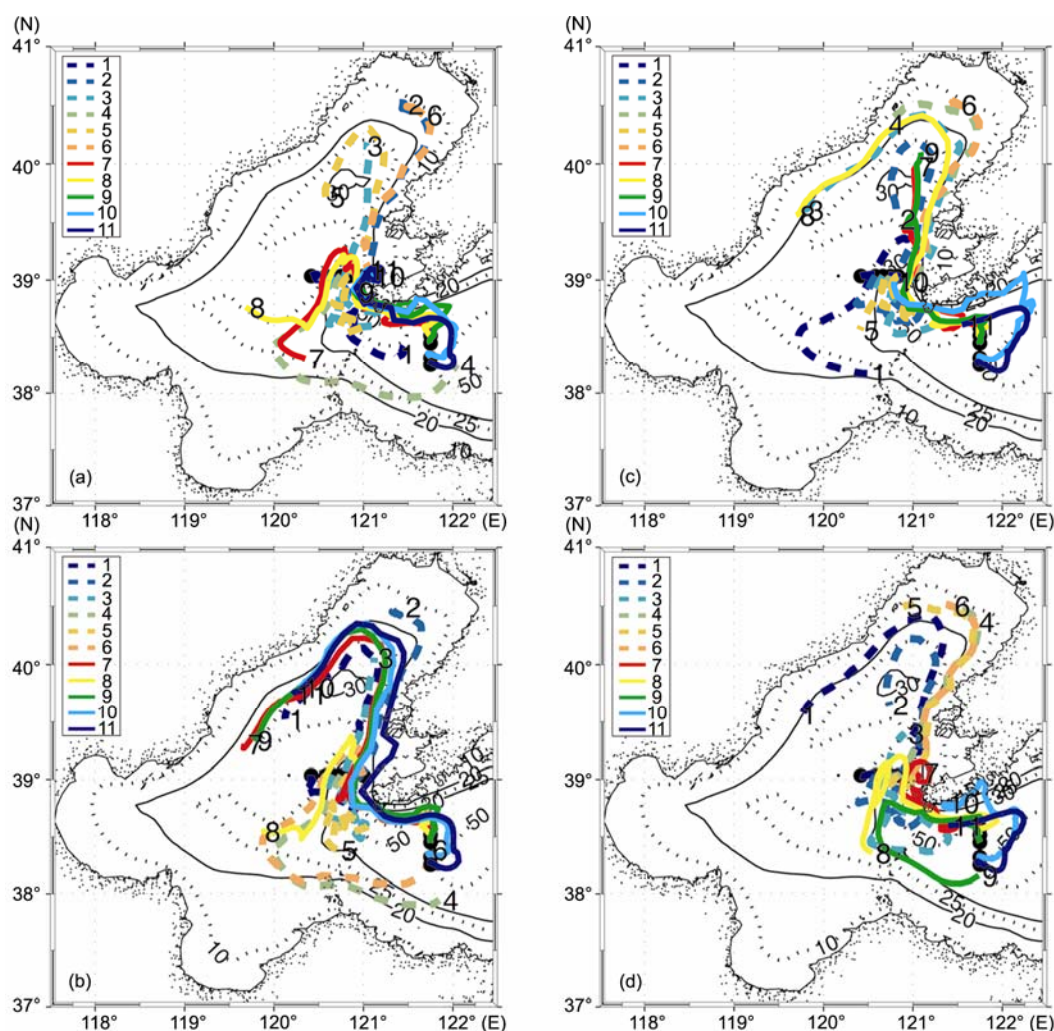


Figure 7 Trajectories of 11 floats according to Lagrangian experiments. The floats are released: (a) from the sub-bottom at flood tides; (b) from the bottom at flood tides; (c) from the sub-bottom at ebb tides; and (d) from the bottom at ebb tides. The floats are named by the number from 1 to 11 which also indicates their ending locations. Their starting sites are denoted by black solid circles at the 39°N and 121°45'E sections. Dashed lines stand for the float trajectories originating from the 39°N section while the others (solid lines) for floats from the 121°45'E section.

float 7 and float 8 turn back to the south direction after reaching beyond the 39°N section and then move southward along the flank of the central bank. In the east of the south depression, float 7 moves eastward and returns to the strait along the 25 m isobath, while float 8 moves westward into southern depression along the same isobath. On the other hand, floats released in the bottom (Figure 7(b)) like 7–9 can enter the northern LDB, while the rest remained in the north of the Laotieshan Channel like float 10 and in the east of the Laotieshan Channel like float 11. Floats 7 and 9 can finally reach the center of the northern depression.

Of floats released in the sub-bottom at the 121°45'E section at ebb tides (Figure 7(c)), all floats except float 8 enter the northern LDB and move anti-clockwise around the depression. Float 8 turns to the southwest after just arriving at the 39°N section and then goes to the east of the southern depression. On the contrary, none of the bottom floats can reach so far as those shallower drifters, which are shown in Figure 7(d). Floats 7, 10 and 11 are trapped to the south of the Liaodong Peninsula, while the rest move anti-clockwise and tend to leave the Bohai Strait.

The 39°N section is located to the north of the Laotieshan Channel and to the south of the north depression. During flooding periods, floats originating from the sub-bottom (Figure 7(a)) like 2, 3, 5 and 6 enter the northern LDB, among which floats 3 and 5 arrive at the edge of the north depression, and floats 2 and 6 near the 10 m isobath. But floats 1 and 4 move southward and go to the northern Yellow Sea. On the other side, floats released from the bottom there (Figure 7(b)) like 2–4 and 6 flow northward into the northern LDB, and of those, float 2 goes through the center of the north depression. But float 1 follows a complicated trajectory. It goes eastward in parallel to the 39°N section at first, then moves northward for a short distance, then turns to the southwest along the flank of the central bank and passes through the east of the south depression, and finally flows eastward to the strait. Float 5 is trapped near the deep channel over most of periods.

At ebb, among the floats starting at the sub-bottom at the section 39°N (Figure 7(c)), floats 1–3 enter the northern LDB and move anti-clockwise. Two of them are around the depression and the other one is along isobaths shallower than 20 m. However, floats 4–6 move essentially southward. The trajectories of floats 4

and 6 are similar to one another. Both of them go northward at first and then turn to southwest along the central bank, pass through the southern depression and move out of the strait. Float 5 spends most of its time in the west of the Laotieshan Channel. By comparison, most of floats originating from the bottom there enter the northern LDB and move anti-clockwise except float 3; float 2 passes through the depression center. Float 3 remains within the Laotieshan Channel (Figure 7(d)).

From the above trajectories of floats both from sub-bottom and bottom, we deduce that the amount and distribution of cold water have a tight relation to their origins. For the floats from the 121°45'E section, some of the bottom floats released at flood tides and some of the sub-bottom floats released at ebb tides can enter the northern LDB. A few of the rest floats reach or pass through the south depression except those staying in the Laotieshan Channel. By comparison, most of the floats from the 39°N section enter the northern LDB and only a small number of them enter or go through the south depression. The floats contributing to the south cold core are mainly from the sub-bottom of the 39°N section, but a few are from the sub-bottom of the northern Yellow Sea. It also implies that cold water in the 39°N section is also contributed by the cold water from the inflow from the northern Yellow Sea and the Laotieshan Channel. According to the floats trajectories, it is suggested that the contribution from the inflow of cold water from the northern Yellow Sea to the two cold core of the Bohai Sea is asymmetric too. The first reason is that the number of floats entering the northern LDB greatly exceeds the number going to the south depression. The second reason is that it takes a relatively long time to travel along the flank of the central bank before their arrival at the south depression, during which they mix with the relatively warm water from the bank.

2.4 Stratification on the central bank

There is no convincing evidence about whether the column on the banks is stratified during summer and, if so how long it can last if any. Table 2 presents some data from historical surveys in July and August. Due to the different sampling techniques and resolutions, we use a simple method to calculate the vertical mean temperature gradient instead of the gradient of each layer to stand for the thermocline intensity. All the records in Table 2 are sampled at the location (39°N, 120°E) close to the transect from MYR to LDB shown in Figure 1.

Table 2 Average temperature gradient on the central bank. (temperature: °C, gradient: °C/m)

	SST (July)	Bottom temperature (July)	Gradient (July)	SST (August)	Bottom temperature (August)	Gradient (August)	Note
Multi-year	23.6	19.0	0.22	25.0	23.0	0.10	ref. [1]
1982	—	—	—	27.6	20.1	0.36	ref. [23]
1992	—	—	—	24.5	24.0	0.02	ref. [23]
1959	23.0	20.0	0.14	—	—	—	ref. [2]
2000	—	—	—	24.5	23.5	0.05	refs. [3,26]
	25.3	18.5	0.34	25.1	22.1	0.15	this study

The average gradient exceeding 0.2 °C/m indicates strong stratification. For example, the SST 23.6°C, the bottom temperature 19.0°C, and the total water depth 21 m in July 1987, resulting in the average gradient 0.22°C/m. But it was only 0.14°C/m in July 1959. In August, only 1982 was a strongly stratified year and the gradient reaches 0.36°C/m while others are not. All the literature cited in Table 2 did not explain the detailed investigation time except for August 2000 (from August 29 to September 5). So the difference in the thermocline intensity might not only be due to the year-to-year variations, but also caused by differences of the observation date. The simulated structure in this paper is in good agreement with the observations of Bao et al. [3].

3 Discussion

Understanding the formation and maintenances of the ADCCBW in stratification seasons can lead to a better understanding of the evolution of the thermocline in the Bohai Sea. The work presented by Lin et al. [7] and this study give clearly suggest that thermocline simulations of the realistic Bohai Sea can't neglect the effects of tidal mixing, especially for the dual-core cold bottom water. Even considering only the M_2 component, it is sufficient to obtain a more reasonable bottom temperature structure and seasonal variations. Lin et al. [7] also take wave effects into consideration to enhance the mixing of upper layers, which deepens the thermocline, weakens its intensity and decreases its duration. Otherwise the thermocline will possibly be overestimated.

However, the well mixed warm water column on the central bank mentioned by Lin et al. [7] does not appear in this paper. The explanation lies in the fact that the most typical summer is July in this inner shelf water when the thermocline is strongest, but August is just a transition time when the thermocline changes very fast. The structure presented above is consistent with the field data observed in late August 2000 mentioned by Bao et al. [3].

Our simulations show that the ADCCBW exists in the two depressions from April to September. In winter, the water is well mixed all over the regions. But there is still 1–2°C disparity between the southern Bohai Sea and the northern LDB besides 2–3°C difference in the two depressions. In general, it is not homogeneous over the whole area even in winter due to the latitude effects and spatial variations of the solar radiation. Moreover, the northern Yellow Sea is much warmer than the Bohai Sea because of the influence from the northward intruding warm water. When spring starts, stratification happens at first at the upper layer above the two depressions. The character of temperature asymmetry remains, especially in the bottom which forms the first stage of the ADCCBW. But the disparity decreases gradually in the following months (Figure 5, right panels) since the mixing effects are dominant than the stratifying in such a shallow region. It is also interesting how the asymmetry retains for such a long term. The present study argues that the inflow of the cold deep water from the northern Yellow Sea and the Laotieshan Channel contribute significantly to maintain the ADCCBW supported by a series of particle tracking experiments.

Our study has focused on the asymmetry of the dual-core structure at the bottom which is an important character in the stratified Bohai Sea in summer. However, since there is always a tight relation between water mass properties and circulations, it is necessary to announce that the present study takes no consideration of water variations during the past periods. Recent works by Wu et al. [26], Lin et al. [27] and Fang et al. [28] have indicate that the average salinity increases by 1–2 PSU in the Bohai Sea during the past decades because of the tremendous reduction of the Yellow River discharge and the increase inflow of the saline Yellow Sea water as well as the rise in the evaporation rate. Annual mean SST increases also at a speed of 0.024°C/a [27]. In spite of that, the ADCCBW structure is pretty obvious in all specific field observations from the 1950s to 2000s as

well as the climatological atlas.

4 Conclusions

The ROMS model driven by climatological atmosphere forcing and boundary forcing is used to simulate stratification in the Bohai Sea, implemented with the $k-\epsilon$ turbulence closure scheme. The model regenerates the stratification processes and reveals that there is a typical ADCCBW structure under the thermocline in the two depression basins, which is surrounding the central bank from April to early September. The bottom temperature in the two depression basins is asymmetric, 1–4°C higher in the south basin.

The imbalance input of the net heat flux caused by the latitude difference and local bathymetry characteristics explain the prior formation of the ADCCBW structure in early stratifications. A series of drifter experiments based on Lagrangian methods suggest that the deep cold water from the northern Yellow Sea and the Laotieshan Channel in the west side of the Liaodong Peninsula contribute to maintaining the structure during late stratification periods. A group of comparison implies that tidal mixing makes the asymmetric structure more significant.

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